

Apriori Algorithm

TID Items

100 1 3 4

200 2 3 5

300 1 2 3 5

400 2 5

C ₁	ItemSet	Min. Sup.	L ₁	ItemSet	Min. Sup.
	{1}	2		{1}	2
	{2}	3		{2}	3
	{3}	3		{3}	3
	{4}	1 → ⊗		{5}	3
	{5}	3			

C ₂	ItemSet	Min. Sup.	L ₂	ItemSet	Min. Sup.
	{1,2}	1 → ⊗		{1,3}	2
	{1,3}	2		{2,3}	2
	{1,5}	1 → ⊗		{2,5}	3
	{2,3}	2		{3,5}	2
	{2,5}	3			
	{3,5}	2			

C ₃	ItemSet	Min. Sup.
	{1,3,2}	1 → ⊗
	{1,3,5}	1 → ⊗
	{2,3,5}	2

Rules	Support	Confidence	%
$2^3 \rightarrow 5$	2	$2/2 = 1$	100%
$3^5 \rightarrow 2$	2	$2/2 = 1$	100%
$5^2 \rightarrow 3$	2	$2/3 = 0.66$	66.6%
$2 \rightarrow 3^5$	2	$2/3 = 0.66$	66.6%
$3 \rightarrow 2^5$	2	$2/3 = 0.66$	66.6%
$5 \rightarrow 2^3$	2	$2/3 = 0.66$	66.6%

2.	TID	Items.
	1	Bread, Milk
	2	Bread, Diapen, Beey, Eggs
	3	Milk, Diapen, Beey, Cola
	4	Milk, Diapen, Beey, Cola
	5	Bread, Milk, Diapen, Cola

Minimum Support = 3

Q ₁	Itemset	Min. Sup	Represent as
	{Bread}	3	B
	{Milk}	4	M
	{Diapen}	4	D
	{Cola}	3	C
	{Eggs}	1	E
	{Beey}	3	Be

L ₁	Itemset	Min. sup.
	{B}	3
	{M}	4
	{D}	4
	{C}	3
	{Be}	3

C ₂	Itemset	Min sup.
	{B, M}	2
	{B, D}	2
	{B, C}	1
	{M, D}	3
	{M, C}	3

$\{D, C\}$	3	
$\{B, A, B, C\}$	1	(X)
$\{M, B, C\}$	2	(X)
$\{D, B, C\}$	3	
$\{C, B, C\}$	2	(X)

L ₂	Itemset	Min sup.
	$\{M, D\}$	3
	$\{M, C\}$	3
	$\{D, C\}$	3
	$\{D, B, C\}$	3

C ₃	Itemset	Min sup.
	$\{M, D, C\}$	3
	$\{M, D, B, C\}$	2 (X)
	$\{D, C, B, C\}$	2 (X)

Association Rules	Support	Confidence	%
$M \rightarrow D \wedge C$	3	$3/4 \rightarrow 0.75$	75%
$D \rightarrow M \wedge C$	3	$3/4 \rightarrow 0.75$	75%
$C \rightarrow M \wedge D$	3	$3/3 \rightarrow 1$	100%
$D \wedge C \rightarrow M$	3	$3/3 \rightarrow 1$	100%
$M \wedge C \rightarrow D$	3	$3/3 \rightarrow 1$	100%
$M \wedge D \rightarrow C$	3	$3/3 \rightarrow 1$	100%